

# ENTERPRISE DATA GOVERNANCE FRAMEWORK DESIGN & IMPLEMENTATION FOR DIGITAL BANKING TRANSFORMATION

## DATA GOVERNANCE TARGET OPERATING MODEL (TOM)

### 1. EXECUTIVE SUMMARY

The Data Governance Target Operating Model (TOM) defines the structure, roles, processes, and technologies required to manage data as a strategic asset across the enterprise. It ensures alignment between business objectives, regulatory requirements, and data management practices.

### 2. OBJECTIVES

- Establish clear accountability for data ownership and stewardship
- Improve data quality and trust across all business domains
- Enable regulatory compliance and auditability
- Standardize data governance processes across the enterprise
- Support scalable governance aligned with modern data platforms

### 3. OPERATING MODEL OVERVIEW

#### 3.1. Governance Model Type

##### ✓ **Federated Governance Model**

- Central governance defines policies & standards
- Domain teams execute governance within business units

##### ✓ **Recommended for Digital Banking because:**

- Multiple business domains
- Strong regulatory oversight
- Shared accountability
- Scalable

## **4. GOVERNANCE STRUCTURE**

### **4.1. Organizational Layers**

#### **1. Level 1: Board Risk Committee**

Responsibilities:

- Oversight
- Risk appetite
- Compliance review

#### **2. Level 2: Chief Data Officer (CDO)**

Responsibilities:

- Enterprise Data Strategy
- Governance leadership
- Regulatory alignment

#### **3. Level 3: Data Governance Council**

Members:

- CDO
- CIO
- CRO
- COO
- Head of Digital Banking
- Head of Compliance

Responsibilities:

- Policy approval
- Issue escalation
- Data standards

#### **4. Level 4: Data Governance Office (DGO)**

Responsibilities:

- Framework execution
- Data quality management
- Metadata management
- Stewardship management

## **5. Level 5: Domain Data Owners**

Domains:

- **Customer Domain**
- **Product Domain**
- **Transaction Domain**
- **Risk Domain**
- **Finance Domain**

Responsibilities:

- Approve definitions
- Data quality accountability
- Issue resolution

## **6. Level 6: Data Stewards**

Responsibilities:

- Data quality monitoring
- Metadata maintenance
- Business rules definition

## **7. Level 7: IT Custodians**

Responsibilities:

- Security controls
- Infrastructure
- Backup & Recovery
- Access management

## 5. RACI FRAMEWORK

| Activity              | Owner | Steward | Custodian | DGO |
|-----------------------|-------|---------|-----------|-----|
| Data Definition       | A     | R       | C         | C   |
| Data Quality Rule     | A     | R       | C         | C   |
| Metadata Update       | C     | R       | C         | A   |
| Data Access Approval  | A     | C       | R         | C   |
| Data Issue Resolution | A     | R       | C         | C   |

(R = Responsible, A = Accountable, C = Consulted, I = Informed)

## 6. CORE GOVERNANCE PROCESSES

### 6.1. Data Quality Management

- Define critical data elements (CDE)

Determination criteria:

- Regulatory/reporting impact
- Revenue/payment impact
- Customer experience impact
- Fraud/risk impact
- Operational dependency across domains

**Critical Data Elements (CDE):**

- Customer ID
- Account Number
- Mobile Number
- Transaction Amount
- Loan Outstanding

### Data Quality Monitoring:

| KPI          | Target |
|--------------|--------|
| Completeness | >95%   |
| Accuracy     | >98%   |
| Timeliness   | >95%   |
| Validity     | >99%   |

- Establish data quality rules & thresholds
- Monitor via dashboards
- Issue management workflow:
  - Detect → Log → Assign → Resolve → Validate → Close

### 6.2. Metadata Management

- Maintain enterprise data catalog
- Define business glossary
- Track data lineage (source → transformation → consumption)

### 6.3. Data Access & Security

- Role-Based Access Control (RBAC)
- Attribute-Based Access Control (ABAC) (optional advanced)
- Access request & approval workflow
- Periodic access review

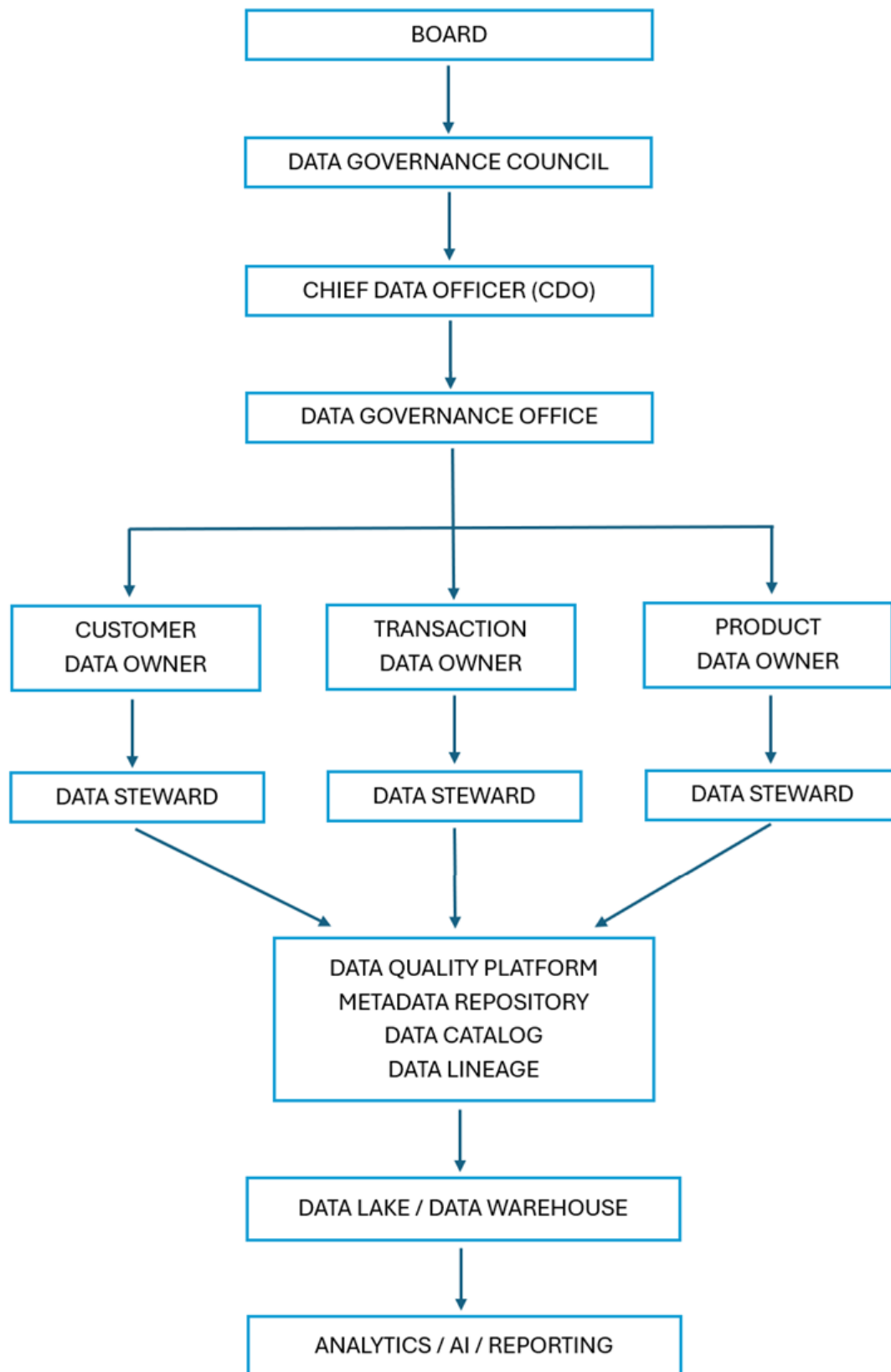
### 6.4. Data Lifecycle Management

- Define lifecycle stages:
  - Create → Store → Use → Archive → Delete
- Retention policies based on regulatory requirements

### 6.5. Data Classification

- Define classification levels:
  - Public
  - Internal
  - Confidential
  - Restricted (PII / highly sensitive)

## 7. DATA GOVERNANCE ARCHITECTURE



## 8. GOVERNANCE FLOW DIAGRAM



## **9. GOVERNANCE ARTEFACTS**

### **Mandatory Artefacts:**

- Data Governance Policy
- Data Standards
- Data Quality Rules
- Data Classification Policy
- Data Access Policy
- Data Dictionary / Glossary
- Data Lineage Documentation
- Governance Control Register

## **10. DATA GOVERNANCE FRAMEWORK COMPONENTS**

### **10.1. Policy Layer**

- Data Governance Policy
- Data Quality Policy
- Metadata Policy
- Data Security Policy

### **10.2. Process Layer**

- Data Ownership
- Data Stewardship
- Data Issue Management
- Data Quality Monitoring

### **10.3. Technology Layer**

- Data Catalog
- Metadata Repository
- DQ Tool
- MDM
- Data Lineage Tool



#### 10.4. People Layer

- CDO
- DGO
- Owners
- Stewards
- Custodians

#### 11. COMPLIANCE & RISK MANAGEMENT

- Align with regulatory frameworks (e.g., GDPR principles)
- Implement audit trails
- Periodic compliance assessment
- Risk identification and mitigation

#### 12. KPIs & SUCCESS METRICS

| KPI                          | Before     | Target              | Success Criteria                |
|------------------------------|------------|---------------------|---------------------------------|
| Data Governance Maturity     | Level 1–2  | Level 4             | Managed & Measurable Governance |
| Data Quality Score           | 82%        | ≥98%                | Trusted Enterprise Data         |
| Metadata Coverage            | 10%        | ≥95%                | Complete Data Documentation     |
| Data Lineage Coverage        | 5%         | ≥90%                | End-to-End Traceability         |
| Regulatory Findings          | 8 Findings | 0 Critical Findings | Regulatory Compliance           |
| Duplicate Customer Records   | 15%        | <2%                 | Single Customer View            |
| Report Reconciliation Effort | 3–5 Days   | <1 Day              | Faster Reporting                |
| Data Issue Resolution SLA    | 20 Days    | ≤10 Days            | Efficient Issue Management      |

#### 13. IMPLEMENTATION ROADMAP

##### Phase 1: Assessment

Deliverables:

- Current State Assessment
- Gap Assessment
- Stakeholder Mapping

## **Phase 2: Framework Design**

Deliverables:

- Governance Model
- Policies
- Standards

## **Phase 3: Technology Enablement**

Deliverables:

- Data Catalog
- DQ Dashboard
- Metadata Repository

## **Phase 4: Pilot Implementation**

Domains:

- Customer
- Transaction

## **Phase 5: Enterprise Rollout**

Deliverables:

- Enterprise Governance
- KPI Dashboard
- Compliance Reporting
- Continuous Improvement

## **14. CHANGE MANAGEMENT & ADOPTION**

- Stakeholder engagement plan
- Training & awareness programs
- Governance champions per domain
- Communication strategy

## 15. ESCALATION FRAMEWORK

- Data issues escalated:
  - Steward → Owner → Governance Council
- SLA for resolution defined

SLA Table (Example):

| Severity | Initial Response | Target Resolution |
|----------|------------------|-------------------|
| Critical | < 4 jam          | < 24 jam          |
| High     | < 1 hari kerja   | < 5 hari kerja    |
| Medium   | < 2 hari kerja   | < 10 hari kerja   |
| Low      | < 5 hari kerja   | < 30 hari kerja   |

## 16. REVIEW & CONTINUOUS IMPROVEMENT

- Quarterly governance review
- Annual TOM update
- Continuous KPI monitoring

## 17. APPENDICES

### Appendix A: Data Quality Dimensions

- Accuracy
- Completeness
- Consistency
- Timeliness

### Appendix B: Sample Workflow

- Data Issue Management Workflow Diagram